

INEQUALITIES

1. INEQUALITIES INVOLVING MODULUS FUNCTION :

Properties of modulus function :

- (i) $|x| \geq a \Rightarrow x \geq a$ or $x \leq -a$, where a is positive.
- (ii) $|x| \leq a \Rightarrow x \in [-a, a]$, where a is positive
- (iii) $|x| > |y| \Rightarrow x^2 > y^2$
- (iv) $||a| - |b|| \leq |a \pm b| \leq |a| + |b|$
- (v) $|x + y| = |x| + |y| \Rightarrow xy \geq 0$
- (vi) $|x - y| = |x| + |y| \Rightarrow xy \leq 0$

Illustration 1 : If x satisfies $|x - 1| + |x - 2| + |x - 3| \geq 6$, then

- (A) $0 \leq x \leq 4$
- (B) $x \leq -2$ or $x \geq 4$
- (C) $x \leq 0$ or $x \geq 4$
- (D) none of these

Solution :

Case I : $x \leq 1$, then

$$1 - x + 2 - x + 3 - x \geq 6 \Rightarrow x \leq 0$$

$$\text{Hence } x \leq 0 \quad \dots(i)$$

Case II : $1 < x \leq 2$, then

$$x - 1 + 2 - x + 3 - x \geq 6 \Rightarrow x \leq -2$$

$$\text{But } 1 < x \leq 2 \Rightarrow \text{No solution} \quad \dots(ii)$$

Case III : $2 < x \leq 3$, then

$$x - 1 + x - 2 + 3 - x \geq 6 \Rightarrow x \geq 6$$

$$\text{But } 2 < x \leq 3 \Rightarrow \text{No solution} \quad \dots(iii)$$

Case IV : $x > 3$, then

$$x - 1 + x - 2 + x - 3 \geq 6 \Rightarrow x \geq 4$$

$$\text{Hence } x \geq 4 \quad \dots(iv)$$

From (i), (ii), (iii) and (iv) the given inequality holds for $x \leq 0$ or $x \geq 4$.

Illustration 2 : Solve for x : (a) $|x - 1| + 2 \leq 4$. (b) $\frac{x-4}{x+2} \leq \left| \frac{x-2}{x-1} \right|$

Solution :

$$(a) |x - 1| + 2 \leq 4 \Rightarrow -4 \leq |x - 1| + 2 \leq 4$$

$$\Rightarrow -6 \leq |x - 1| \leq 2$$

$$\Rightarrow |x - 1| \leq 2 \Rightarrow -2 \leq x - 1 \leq 2$$

$$\Rightarrow -1 \leq x \leq 3 \Rightarrow x \in [-1, 3]$$

(b) **Case 1 :** Given inequation will be satisfied for all x such that

$$\frac{x-4}{x+2} \leq 0 \Rightarrow x \in (-2, 4] - \{1\} \quad \dots(i)$$

(Note : $\{1\}$ is not in domain of RHS)

$$\text{Case 2 : } \frac{x-4}{x+2} > 0 \Rightarrow x \in (-\infty, -2) \cup (4, \infty) \dots\dots(ii)$$

Given inequation becomes

$$\frac{x-2}{x-1} \geq \frac{x-4}{x+2} \quad \text{or} \quad \frac{x-2}{x-1} \leq -\frac{x-4}{x+2}$$

on solving we get

$$x \in (-2, 4/5) \cup (1, \infty)$$

on solving we get

$$x \in (-2, 0] \cup (1, 5/2]$$

taking intersection with (ii) we get taking intersection with (ii) we get

$$x \in (4, \infty)$$

.....(iii)

$$x \in \phi$$

Hence, solution of the original inequation: $x \in (-2, \infty) - \{1\}$ (taking union of (i) & (iii))

Illustration 3 : The equation $|x| + \frac{x}{x-1} = \frac{x^2}{x-1}$ is always true for x belongs to

$$(A) \{0\}$$

$$(B) (1, \infty)$$

$$(C) (-1, 1)$$

$$(D) (-\infty, \infty)$$

Solution :

$$\frac{x^2}{x-1} = \left| x + \frac{x}{x-1} \right|$$

$$\therefore |x| + \frac{x}{x-1} = \left| x + \frac{x}{x-1} \right| \text{ is true only if } \left(x \cdot \frac{x}{x-1} \right) \geq 0 \Rightarrow x \in \{0\} \cup (1, \infty) \text{ Ans. (A,B)}$$

2. IRRATIONAL INEQUALITIES :

Illustration 4 : Solve for x, if $\sqrt{x^2 - 3x + 2} > x - 2$

Solution :

$$\left[\begin{array}{l} \left\{ \begin{array}{l} x^2 - 3x + 2 \geq 0 \\ x - 2 \geq 0 \end{array} \right. \\ (x^2 - 3x + 2) > (x - 2)^2 \end{array} \right. \quad \text{or} \quad \Rightarrow \quad \left[\begin{array}{l} \left\{ \begin{array}{l} (x-1)(x-2) \geq 0 \\ (x-2) \geq 0 \Rightarrow x > 2 \\ x - 2 > 0 \end{array} \right. \\ \text{or} \\ \left\{ \begin{array}{l} (x-1)(x-2) \geq 0 \\ x - 2 < 0 \end{array} \right. \Rightarrow x \leq 1 \end{array} \right.$$

Hence, solution set of the original inequation is $x \in \mathbb{R} - (1, 2]$

Do yourself - 1 :

(i) Solve for x if $\frac{|x^2 - 4|}{x^2 + x - 2} > 1$

(ii) Solve for x if $\sqrt{x^2 - x} > (x - 1)$

3. LOGARITHMIC INEQUALITIES :

Points to remember :

- (i) $\log_a x < \log_a y \Leftrightarrow \begin{cases} x < y & \text{if } a > 1 \\ x > y & \text{if } 0 < a < 1 \end{cases}$
- (ii) If $a > 1$, then (a) $\log_a x < p \Rightarrow 0 < x < a^p$ (b) $\log_a x > p \Rightarrow x > a^p$
- (iii) If $0 < a < 1$, then (a) $\log_a x < p \Rightarrow x > a^p$ (b) $\log_a x > p \Rightarrow 0 < x < a^p$

Illustration 5 : Solve for x : (a) $\log_{0.5} (x^2 - 5x + 6) \geq -1$ (b) $\log_{1/3} (\log_4 (x^2 - 5)) > 0$

Solution : (a) $\log_{0.5} (x^2 - 5x + 6) > -1 \Rightarrow 0 < x^2 - 5x + 6 \leq (0.5)^{-1}$
 $\Rightarrow 0 < x^2 - 5x + 6 \leq 2$

$$\begin{cases} x^2 - 5x + 6 > 0 \\ x^2 - 5x + 6 \leq 2 \end{cases} \Rightarrow x \in [1, 2) \cup (3, 4]$$

Hence, solution set of original inequation : $x \in [1, 2) \cup (3, 4]$

(b) $\log_{1/3} (\log_4 (x^2 - 5)) > 0 \Rightarrow 0 < \log_4 (x^2 - 5) < 1$

$$\begin{cases} 0 < \log_4 (x^2 - 5) \Rightarrow x^2 - 5 > 1 \\ \log_4 (x^2 - 5) < 1 \Rightarrow 0 < x^2 - 5 < 4 \end{cases} \Rightarrow 1 < (x^2 - 5) < 4$$

$$\Rightarrow 6 < x^2 < 9$$

$$\Rightarrow x \in (-3, -\sqrt{6}) \cup (\sqrt{6}, 3)$$

Hence, solution set of original inequation : $x \in (-3, -\sqrt{6}) \cup (\sqrt{6}, 3)$

Illustration 6 : Solve for x : $\log_2 x \leq \frac{2}{\log_2 x - 1}$

Solution : Let $\log_2 x = t$

$$t \leq \frac{2}{t-1} \Rightarrow t - \frac{2}{t-1} \leq 0$$

$$\Rightarrow \frac{t^2 - t - 2}{t-1} \leq 0 \Rightarrow \frac{(t-2)(t+1)}{(t-1)} \leq 0 \Rightarrow t \in (-\infty, -1] \cup (1, 2]$$

$$\text{or } \log_2 x \in (-\infty, -1] \cup (1, 2]$$

$$\text{or } x \in \left(0, \frac{1}{2}\right] \cup (2, 4]$$

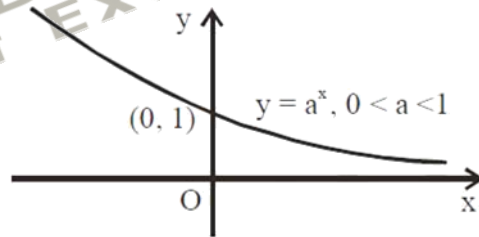
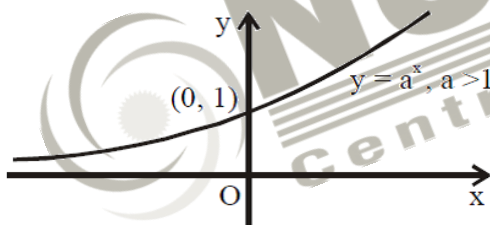
Illustration 7 : Solve the inequation : $\log_{2x+3} x^2 < \log_{2x+3} (2x + 3)$

Solution : This inequation is equivalent to the collection of the systems

$$\left[\begin{array}{l} \left\{ \begin{array}{l} 2x+3 > 1 \\ 0 < x^2 < 2x+3 \end{array} \right. \\ \text{or} \\ \left\{ \begin{array}{l} 0 < 2x+3 < 1 \\ x^2 > 2x+3 > 0 \end{array} \right. \end{array} \right] \Rightarrow \left[\begin{array}{l} \left\{ \begin{array}{l} x > -1 \\ (x-3)(x+1) < 0 \text{ \& } x \neq 0 \end{array} \right. \\ \text{or} \\ \left\{ \begin{array}{l} -\frac{3}{2} < x < -1 \\ (x-3)(x+1) > 0 \end{array} \right. \end{array} \right] \Rightarrow \left[\begin{array}{l} \left\{ \begin{array}{l} x > -1 \\ -1 < x < 3 \end{array} \right. \Rightarrow -1 < x < 3 \text{ \& } x \neq 0 \\ \text{or} \\ \left\{ \begin{array}{l} -\frac{3}{2} < x < -1 \\ x < -1 \text{ or } x > 3 \end{array} \right. \Rightarrow -\frac{3}{2} < x < -1 \end{array} \right]$$

Hence, solution of the original inequation is $x \in \left(-\frac{3}{2}, -1\right) \cup (-1, 0) \cup (0, 3)$

4. EXPONENTIAL INEQUATIONS :



$$\text{If } a^{f(x)} > b \Rightarrow \begin{cases} f(x) > \log_a b \text{ when } a > 1 \\ f(x) < \log_a b \text{ when } 0 < a < 1 \end{cases}$$

Illustration 8 : Solve for x : $2^{x+2} > \left(\frac{1}{4}\right)^{\frac{1}{x}}$

Solution : We have $2^{x+2} > 2^{-2/x}$. Since the base $2 > 1$, we have $x + 2 > -\frac{2}{x}$

(the sign of the inequality is retained).

$$\text{Now } x + 2 + \frac{2}{x} > 0 \Rightarrow \frac{x^2 + 2x + 2}{x} > 0 \Rightarrow \frac{(x+1)^2 + 1}{x} > 0 \Rightarrow x \in (0, \infty)$$

Illustration 9 : Solve for x : $(1.25)^{1-x} < (0.64)^{2(1+\sqrt{x})}$

Solution : We have $\left(\frac{5}{4}\right)^{1-x} < \left(\frac{16}{25}\right)^{2(1+\sqrt{x})}$ or $\left(\frac{4}{5}\right)^{x-1} < \left(\frac{4}{5}\right)^{4(1+\sqrt{x})}$

Since the base $0 < \frac{4}{5} < 1$, the inequality is equivalent to the inequality

$$x - 1 > 4(1 + \sqrt{x}) \Rightarrow \frac{x-5}{4} > \sqrt{x}$$

Now, R.H.S. is positive

$$\Rightarrow \frac{x-5}{4} > 0 \Rightarrow x > 5 \quad \dots\dots(i)$$

we have $\frac{x-5}{4} > \sqrt{x}$ both sides are positive, so squaring both sides

$$\Rightarrow \frac{(x-5)^2}{16} > x \quad \text{or} \quad \frac{(x-5)^2}{16} - x > 0$$

$$\text{or } x^2 - 26x + 25 > 0 \quad \text{or } (x-25)(x-1) > 0$$

$$\Rightarrow x \in (-\infty, 1) \cup (25, \infty) \quad \dots\dots(ii)$$

intersection (i) & (ii) gives $x \in (25, \infty)$

Do yourself - 2 :

(i) Solve for x : (a) $\log_{0.3}(x^2 + 8) > \log_{0.3}(9x)$ (b) $\log_7\left(\frac{2x-6}{2x-1}\right) > 0$

(ii) Solve for x : $\left(\frac{2}{3}\right)^{\frac{|x|-1}{|x|+1}} > 1$.

INEQUALITIES BASED ON RMS, AM, GM & HM

Note : Given $a_1, a_2, a_3, \dots, a_n$ are positive numbers.

$$\text{Root Mean Square, RMS} = \sqrt{\frac{a_1^2 + a_2^2 + \dots + a_n^2}{n}}$$

$$\text{Arithmetic Mean, AM} = \frac{a_1 + a_2 + a_3 + \dots + a_n}{n}$$

$$\text{Geometric Mean, GM} = (a_1 a_2 a_3 \dots a_n)^{1/n}$$

$$\text{Harmonic Mean, HM} = \frac{n}{\frac{1}{a_1} + \frac{1}{a_2} + \dots + \frac{1}{a_n}}$$

We have a relation,

$$\text{RMS} \geq \text{AM} \geq \text{GM} \geq \text{HM}$$

Equality holds when $a_1 = a_2 = a_3 \dots = a_n$

Illustration 10 : If a, b are the positive numbers & $3a + b = 5$ then find maximum value of

- (i) a^3b (ii) a^2b (iii) ab^3

Also explain holding of equality

Solution :

$$(i) \frac{a + a + a + b}{4} \geq (a \cdot a \cdot a \cdot b)^{1/4}$$

$$\Rightarrow \frac{3a + b}{4} \geq (a^3b)^{1/4} \Rightarrow \frac{5^4}{4^4} \geq a^3b$$

Equality hold when $a = a = a = b$

$$\& 3a + b = 5 \Rightarrow a = \frac{5}{4} = b$$

$$(ii) \frac{\frac{3a}{2} + \frac{3a}{2} + b}{3} \geq \left(\frac{3a}{2} \cdot \frac{3a}{2} \cdot b \right)^{1/3}$$

$$\Rightarrow \frac{3a + b}{3} \geq \left(\frac{9}{4} a^2b \right)^{1/3} \Rightarrow \left(\frac{5}{3} \right)^3 \geq \frac{9}{4} a^2b$$

$$\Rightarrow \frac{500}{243} \geq a^2b.$$

Equality hold when $\frac{3a}{2} = b$

$$3a + b = 5 \Rightarrow 3b = 5 \Rightarrow b = \frac{5}{3} \& a = \frac{10}{9}$$

$$(iii) \frac{3a + \frac{b}{3} + \frac{b}{3} + \frac{b}{3}}{4} \geq \left(3a \cdot \frac{b}{3} \cdot \frac{b}{3} \cdot \frac{b}{3} \right)^{1/4}$$

$$\Rightarrow \frac{3a + b}{4} \geq \left(\frac{ab^3}{2^2} \right)^{1/4} \Rightarrow \left(\frac{5}{4} \right)^4 \times 3^2 \geq ab^3$$

Illustration 11 : If $a, b, c \in \mathbb{R}^+$ Prove that $\frac{a^3}{4b^2} + \frac{b}{2c} + \frac{81c^2}{4a^3} \geq 3$

Solution :

$$\frac{\frac{a^3}{4b^2} + \frac{b}{4c} + \frac{b}{4c} + \frac{81c^2}{4a^3}}{4} \geq \left(\frac{a^3}{4b^2} \times \frac{b}{4c} \times \frac{b}{4c} \times \frac{81c^2}{4a^3} \right)^{1/4}$$

$$\Rightarrow \frac{a^3}{2b} + \frac{b}{2c} + \frac{81c^2}{4a^3} \geq 4 \times \left(\frac{81}{4^4} \right)^{1/4}$$

$$\Rightarrow \frac{a^3}{2b} + \frac{b}{2c} + \frac{81c^2}{4a^3} \geq (81)^{1/4} = 3$$

Illustration 12 : $a > 0$; $b > 0$ and $a + b = 1$, show that $\left(a + \frac{1}{a}\right)^2 + \left(b + \frac{1}{b}\right)^2 \geq 12.5$

Solution : Let $a = \sin^2\theta$ & $b = \cos^2\theta$

$$\begin{aligned}\therefore \left(a + \frac{1}{a}\right)^2 + \left(b + \frac{1}{b}\right)^2 &= \sin^4\theta + \cos^4\theta + \sec^4\theta + \operatorname{cosec}^4\theta + 4 \\ &= (\sin^2\theta + \cos^2\theta) - 2\sin^2\theta\cos^2\theta + (1 + \tan^2\theta)^2 + (1 + \cot^2\theta)^2 + 4 \\ &= 1 - 2\sin^2\theta\cos^2\theta + \tan^4\theta + \cot^4\theta + 2(\tan^2\theta + \cot^2\theta) + 6. \\ &= 7 + (\tan^4\theta + \cot^4\theta) + 2(\tan^2\theta + \cot^2\theta) - \frac{1}{2}\sin^22\theta\end{aligned}$$

Note : $0 \leq \frac{1}{2}\sin^22\theta \leq \frac{1}{2} \Rightarrow -\frac{1}{2}\sin^2\theta \geq -\frac{1}{2}$ [Equality at $\sin2\theta = 1 \Rightarrow \theta = 45^\circ$]

$$\tan^4\theta + \cos^4\theta \geq 2\sqrt{\tan^4\theta \cdot \cot^4\theta} \Rightarrow \tan^4\theta + \cot^4\theta \geq 2 \text{ [Equality at } \theta = 45^\circ]$$

$$\& \tan^2\theta + \cot^2\theta \geq 2 \text{ [Equality at } \theta = 45^\circ]$$

$$\text{So, } \left(a + \frac{1}{a}\right)^2 + \left(b + \frac{1}{b}\right)^2 \geq 7 + 2 + 2 \times 2 - \frac{1}{2}$$

$$\Rightarrow \left(a + \frac{1}{a}\right)^2 + \left(b + \frac{1}{b}\right)^2 \geq 12.5$$

Illustration 13 : If a, b, x, y are positive rational numbers such that $\frac{1}{x} + \frac{1}{y} = 1$,

then prove that $\frac{a^x}{x} + \frac{b^y}{y} \geq ab$

Solution:

Let LCM of denominators of x & y be k then kx & ky will be positive integers

Let $a^x, a^x, a^x \dots ky$ times & $b^y, b^y, b^y, \dots kx$ times

$$\text{So AM} = \frac{(a^x + a^x + \dots (ky \text{ times})) + (b^y + b^y + \dots (kx \text{ times}))}{kx + ky}$$

$$= \frac{a^x \cdot ky + b^y \cdot kx}{kx + ky} = \frac{a^x \cdot y + b^y \cdot x}{x + y}$$

$$\text{GM} = \left((a^x)^{ky} \cdot (b^y)^{kx}\right)^{\frac{1}{kx+ky}}$$

$$\text{GM} = (ab)^{\frac{xy}{x+y}} = ab \quad \left\{ \frac{1}{x} + \frac{1}{y} = 1 \Rightarrow \frac{x+y}{xy} = 1 \right\}$$

Now $AM \geq GM$

$$\therefore \frac{a^x y + b^y x}{x + y} \geq ab \quad [\text{But } x + y = xy]$$

$$\text{So, } \frac{a^x y + b^y x}{xy} \geq ab$$

$$\Rightarrow \frac{a^x}{x} + \frac{b^y}{y} \geq ab \quad [\text{Hence Proved}]$$

Aliter :

$$x = \frac{p}{q_1} \text{ \& \; } y = \frac{p}{q_2} ; \frac{1}{x} + \frac{1}{y} = 1 \quad [p = q_1 + q_2] ; p, q_1, q_2 \in I$$

$$\text{LHS} = \frac{a^x}{x} + \frac{b^y}{y} = \frac{(a^{\frac{p}{q_1}})q_1}{p} + \frac{(b^{\frac{p}{q_2}})q_2}{p} = \frac{(a^{\frac{p}{q_1}})q_1 + (b^{\frac{p}{q_2}})q_2}{q_1 + q_2}$$

$$\text{LHS} \geq \left((a^{\frac{p}{q_1}})^{q_1} (b^{\frac{p}{q_2}})^{q_2} \right)^{\frac{1}{q_1 + q_2}} \quad (\text{AM} \geq \text{GM})$$

Illustration 14 : If $2x^3 + ax^2 + bx + 4 = 0$ (a and b are positive real numbers) has 3 real roots, then prove that $a + b \geq 6(2^{1/3} + 4^{1/3})$.

Solution : Let α, β, γ be the roots of $2x^3 + ax^2 + bx + 4 = 0$. Given that all the coefficients are positive, so all the roots will be negative.

$$\text{Let } \alpha_1 = -\alpha, \alpha_2 = -\beta, \alpha_3 = -\gamma \quad \Rightarrow \alpha_1 + \alpha_2 + \alpha_3 = \frac{a}{2}$$

$$\alpha_1 \alpha_2 + \alpha_2 \alpha_3 + \alpha_3 \alpha_1 = \frac{b}{2}$$

$$\alpha_1 \alpha_2 \alpha_3 = 2$$

Applying $AM \geq GM$, we have

$$\frac{\alpha_1 \alpha_2 \alpha_3}{3} \geq (\alpha_1 \alpha_2 \alpha_3)^{1/3}$$

$$\Rightarrow a \geq 6 \times 2^{1/3}$$

$$\text{Also } \frac{\alpha_1 \alpha_2 + \alpha_2 \alpha_3 + \alpha_3 \alpha_1}{3} > (\alpha_1 \alpha_2 \alpha_3)^{2/3} \Rightarrow b \geq 6 \times 4^{1/3}$$

Therefore $a + b \geq 6(2^{1/3} + 4^{1/3})$.

Illustration 15 : If $a_i > 0 \forall i \in \mathbb{N}$ such that $\prod_{i=1}^n a_i = 1$, then prove that

$$(1 + a_1)(1 + a_2)(1 + a_3).....(1 + a_n) \geq 2^n$$

Solution : Using A.M. $\geq$ G.M.

$$1 + a_1 \geq 2\sqrt{a_1}$$

$$1 + a_2 \geq 2\sqrt{a_2}$$

$$1 + a_n \geq 2\sqrt{a_n} \Rightarrow (1 + a_1)(1 + a_2).....(1 + a_n) \geq 2^n(a_1 a_2 a_3 a_n)^{1/2}$$

$$\text{As } a_1 a_2 a_3 a_n = 1$$

$$\text{Hence } (1 + a_1)(1 + a_2).....(1 + a_n) \geq 2^n.$$

Do yourself - 3 :

(i) Find minimum value of

(a) $8\sec^2\theta + 18\operatorname{cosec}^2\theta$

(b) $\tan^2\theta + 4\cot^2\theta$

(ii) If $0 < r < 1$ and m is a positive integer, show that $(2m + 1)r^m(1 - r) < 1 - r^{2m+1}$.

(iii) If $a > 0, b > 0, c > 0$ prove that $\frac{1}{s-a} + \frac{1}{s-b} + \frac{1}{s-c} > \frac{9}{2s}$, where $s = a + b + c$

(iv) (a) If $\frac{a^n + b^n}{a^{n-1} + b^{n-1}}$ is the G.M. between a & b then find the value of 'n'.

(b) If b is the harmonic mean between a and c , then prove that $\frac{1}{b-a} + \frac{1}{b-c} = \frac{1}{a} + \frac{1}{c}$.

CAUCHY SCHWARTZ INEQUALITY

If $a_1, a_2, \dots, a_n, b_1, b_2, b_3, \dots, b_n$ are real numbers then

$$(a_1^2 + a_2^2 + a_3^2 + \dots + a_n^2) \cdot (b_1^2 + b_2^2 + b_3^2 + \dots + b_n^2) \geq (a_1 b_1 + a_2 b_2 + a_3 b_3 + \dots + a_n b_n)^2$$

Proof : It comes from a result of quadratic expression.

We know that

$$(a_1 x - b_1)^2 + (a_2 x - b_2)^2 + (a_3 x - b_3)^2 + \dots + (a_n x - b_n)^2 \geq 0 \forall x \in \mathbb{R}$$

$$\Rightarrow (a_1^2 + a_2^2 + a_3^2 + \dots + a_n^2)x^2 - 2(a_1 b_1 + a_2 b_2 + a_3 b_3 + \dots + a_n b_n)x + (b_1^2 + b_2^2 + b_3^2 + \dots + b_n^2) \geq 0 \forall x \in \mathbb{R}$$

Since coefficient of $x^2 > 0$ then $D \leq 0$

$$\Rightarrow 4(a_1 b_1 + a_2 b_2 + \dots + a_n b_n)^2 - 4(a_1^2 + a_2^2 + a_3^2 + \dots + a_n^2)(b_1^2 + b_2^2 + \dots + b_n^2) \leq 0.$$

$$\Rightarrow \left(\sum_{i=1}^n a_i b_i \right)^2 \leq \left(\sum_{i=1}^n a_i^2 \right) \left(\sum_{i=1}^n b_i^2 \right) \quad [\text{Hence Proved}]$$

$$\text{Equality holds when } \frac{a_1}{b_1} = \frac{a_2}{b_2} = \frac{a_3}{b_3} = \dots = \frac{a_n}{b_n} = x$$

Proof of Cauchy Schwartz inequality by vectors (for two sets of three variables)

Let $\vec{a} = a_1\hat{i} + a_2\hat{j} + a_3\hat{k}$ & $\vec{b} = b_1\hat{i} + b_2\hat{j} + b_3\hat{k}$

Where $\hat{i}, \hat{j}$ & $\hat{k}$ are unit vector along x, y & z respectively.

$$\vec{a} \cdot \vec{b} = |\vec{a}| |\vec{b}| \cos \theta \leq |\vec{a}| |\vec{b}|$$

[as $\cos \theta \leq 1$, equality holds when $\theta = 0^\circ$ means vectors are parallel]

$$\Rightarrow (\vec{a} \cdot \vec{b})^2 \leq |\vec{a}|^2 |\vec{b}|^2$$

$$\Rightarrow (a_1b_1 + a_2b_2 + a_3b_3)^2 \leq (a_1^2 + a_2^2 + a_3^2)(b_1^2 + b_2^2 + b_3^2)$$

Lemma (short theorem) on Cauchy – Schwartz Inequality

Let $a_1, a_2, \dots, a_n \in \mathbb{R}$ & $x_1, x_2, x_3, \dots, x_n \in \mathbb{R}^+$

Then
$$\left(\frac{a_1^2}{x_1} + \frac{a_2^2}{x_2} + \frac{a_3^2}{x_3} + \dots + \frac{a_n^2}{x_n} \right) \geq \frac{(a_1 + a_2 + \dots + a_n)^2}{(x_1 + x_2 + x_3 + \dots + x_n)}$$

Proof
$$\left[\left(\frac{a_1}{\sqrt{x_1}} \right)^2 + \left(\frac{a_2}{\sqrt{x_2}} \right)^2 + \left(\frac{a_3}{\sqrt{x_3}} \right)^2 + \dots + \left(\frac{a_n}{\sqrt{x_n}} \right)^2 \right] \cdot \left[(\sqrt{x_1})^2 + (\sqrt{x_2})^2 + \dots + (\sqrt{x_n})^2 \right]$$

$$\geq (a_1 + a_2 + \dots + a_n)^2$$

$$\Rightarrow \left(\frac{a_1^2}{x_1} + \frac{a_2^2}{x_2} + \dots + \frac{a_n^2}{x_n} \right) \geq \frac{(a_1 + a_2 + a_3 + \dots + a_n)^2}{(x_1 + x_2 + x_3 + \dots + x_n)}$$

Equality holds when $\frac{\frac{a_1}{\sqrt{x_1}}}{\sqrt{x_1}} = \frac{a_1}{x_1} = \frac{a_2}{x_2} = \frac{a_3}{x_3} = \dots = \frac{a_n}{x_n}$

Illustration 16 : Let $a, b, c \in \mathbb{R}^+$ prove that $\frac{a}{b+c} + \frac{b}{c+a} + \frac{c}{a+b} \geq \frac{3}{2}$

Solution :
$$\frac{a}{b+c} + \frac{b}{c+a} + \frac{c}{a+b} = \frac{a^2}{ab+ac} + \frac{b^2}{bc+ab} + \frac{c^2}{ab+bc} \geq \frac{(a+b+c)^2}{2(ab+bc+ca)}$$

[Using lemma of C - Shz]

$$\Rightarrow \frac{a}{b+c} + \frac{b}{c+a} + \frac{c}{a+b} \geq \frac{(a+b+c)^2}{2(ab+bc+ca)}$$

We know that, $a^2 + b^2 + c^2 \geq ab + bc + ca$

$$\Rightarrow \frac{1}{2}[(a-b)^2 + (b-c)^2 + (c-a)^2] \geq 0 \Rightarrow \frac{(a+b+c)^2}{ab+bc+ca} \geq 3$$

Hence; $\frac{a}{b+c} + \frac{b}{c+a} + \frac{c}{a+b} \geq \frac{3}{2}$.

Illustration 17 : Let $a, b \in \mathbb{R}^+$, Prove that $8(a^4 + b^4) \geq (a + b)^4$

Solution :
$$8(a^4 + b^4) = 8 \left[\frac{(a^2)^2}{1} + \frac{(b^2)^2}{1} \right] \geq 8 \frac{(a^2 + b^2)^2}{(1+1)} = 4(a^2 + b^2)^2$$

Now,
$$4(a^2 + b^2) = 4 \left(\frac{a^2}{1} + \frac{b^2}{1} \right) \geq 4 \left(\frac{(a+b)^2}{(1+1)} \right)^2 = (a+b)^4.$$

[This can also be prove by $\text{RMS} \leq \text{AM}$]

[Using Lemma of Cauchy – Schwartz] [Hence Proved].

Do yourself - 4 :

(i) Let $x, y, z > 0$. Prove that
$$\frac{2}{x+y} + \frac{2}{y+z} + \frac{2}{z+x} \geq \frac{9}{x+y+z}$$

(ii) Find the minimum value of $\sqrt{a^2 + b^2}$ when $3a + 4b = 15$.

JENSEN'S INEQUALITY

Suppose $f(x)$ is a twice differentiable function on an interval $[a, b]$ and $f''(x) < 0 \quad \forall a < x < b$.

Then for every +ve integer m and for all points.

$x_1, x_2, x_3, \dots, x_m$ in $[a, b]$, we have

$$f\left(\frac{x_1 + x_2 + x_3 + \dots + x_m}{m}\right) \geq \frac{f(x_1) + f(x_2) + \dots + f(x_m)}{m}$$

Moreover equality holds if and only if $x_1 = x_2 = x_3 = \dots, x_n$

Note : in this case, graph of $f(x)$ is concave down.

Corollary : Let $f''(x) < 0$

and let $x_1, x_2 \in [a, b]$, then

$$f\left(\frac{x_1 + x_2}{2}\right) > \frac{f(x_1) + f(x_2)}{2}$$

Coordinates of M are $\left\{ \frac{x_1 + x_2}{2}, \frac{f(x_1) + f(x_2)}{2} \right\}$

Coordinates of R are $\left\{ \frac{x_1 + x_2}{2}, f\left(\frac{x_1 + x_2}{2}\right) \right\}$

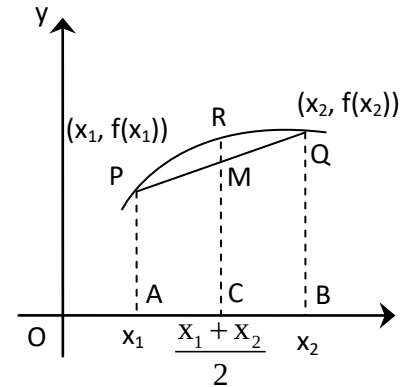
Now, from the figure

Algebraic value of RC > Algebraic value of MC

$$RC = f\left(\frac{x_1 + x_2}{2}\right)$$

$$MC = \frac{f(x_1) + f(x_2)}{2}$$

$$f\left(\frac{x_1 + x_2}{2}\right) \geq \frac{f(x_1) + f(x_2)}{2}$$



Concept : If $f''(x) > 0, \forall x \in]a, b[$, then

$$f\left(\frac{x_1 + x_2 + \dots + x_m}{m}\right) \geq \frac{f(x_1) + f(x_2) + \dots + f(x_m)}{m}$$

Note : in this case, graph of $f(x)$ is concave up.

It is easy to derive the AM–GM inequality as a special case of the Jensen's inequality.

Suppose $y_1, y_2, \dots, y_m$ are positive real numbers.

Let $x_i = \log y_i$ for $i = 1, 2, \dots, m$

i.e., logarithm w.r.t. the base e (so that for $i = 1, 2, \dots, m$, we have $y_i = e^{x_i}$)

Take function $h(x) = e^x$ satisfies the condition in the inequality.

$\therefore h''(x) = e^x$ is +ve.

$\therefore$ By Jensen's inequality, we have

$$e^{\left(\frac{x_1 + x_2 + \dots + x_m}{m}\right)} \geq \frac{e^{x_1} + e^{x_2} + \dots + e^{x_m}}{m}$$

$\therefore e^{x_i} = y_i$ for $i = 1, 2, \dots, m$

$$\text{i.e., } (y_1 y_2 \dots y_m)^{\frac{1}{m}} \geq \frac{y_1 + y_2 + y_m + \dots}{m}$$

Which is AM–GM inequality. Also equality holds if and only if all x_i 's are equal which is same thing as saying that all y_i 's are equal.

Illustration 18 : ABC is an acute angled triangle show that

$$\cos A + \cos B + \cos C \leq 3/2$$

Solution : $\therefore$ ABC is an acute angled triangle.

$\therefore$ A, B, C would lie in the interval $[0, \pi/2]$

$$\text{Let } h(x) = \cos x$$

$$h'(x) = -\sin x$$

$$h''(x) = -\cos x < 0 \quad \forall x \in \left(0, \frac{\pi}{2}\right)$$

By using Jensen's inequality

$$f\left(\frac{x_1 + x_2 + \dots + x_m}{m}\right) \geq \frac{f(x_1) + f(x_2) + \dots + f(x_m)}{m}$$

$$\Rightarrow \cos\left(\frac{A+B+C}{3}\right) \geq \frac{\cos A + \cos B + \cos C}{3}$$

Now, $A + B + C = \pi$

$$\Rightarrow \cos \frac{\pi}{3} \geq \frac{\cos A + \cos B + \cos C}{3}$$

$$\Rightarrow \cos A + \cos B + \cos C \leq 3/2$$

Equality holds if and only if $A = B = C$

Do yourself -5 :

(i) If $0 < A_i < \pi, \forall i = 1, 2, \dots, n$, show that

$$\sin A_1 + \sin A_2 + \dots + \sin A_n \geq n \sin \left(\frac{A_1 + A_2 + \dots + A_n}{n} \right)$$

MAXIMUM & MINIMUM VALUES OF TRIGONOMETRIC EXPRESSIONS :

- (i) $a \cos \theta + b \sin \theta$ will always lie in the interval $[-\sqrt{a^2 + b^2}, \sqrt{a^2 + b^2}]$ i.e. the maximum and minimum values are $\sqrt{a^2 + b^2}, -\sqrt{a^2 + b^2}$ respectively.
- (ii) Minimum value of $a^2 \tan^2 \theta + b^2 \cot^2 \theta = 2ab$ where $a, b > 0$
- (iii) $-\sqrt{a^2 + b^2 + 2ab \cos(\alpha - \beta)} \leq a \cos(\alpha + \theta) + b \cos(\beta + \theta) \leq \sqrt{a^2 + b^2 + 2ab \cos(\alpha - \beta)}$ where α and β are known angles.
- (iv) In case a quadratic in $\sin \theta$ & $\cos \theta$ is given then the maximum or minimum values can be obtained by making perfect square.

Illustration 19 : Prove that : $-4 \leq 5 \cos \theta + 3 \cos \left(\theta + \frac{\pi}{3} \right) + 3 \leq 10$, for all values of θ .

Solution :

We have,

$$5 \cos \theta + 3 \cos \left(\theta + \frac{\pi}{3} \right) = 5 \cos \theta + 3 \cos \theta \cos \frac{\pi}{3} - 3 \sin \theta \sin \frac{\pi}{3} = \frac{13}{2} \cos \theta - \frac{3\sqrt{3}}{2} \sin \theta$$

$$\text{Since, } -\sqrt{\left(\frac{13}{2}\right)^2 + \left(-\frac{3\sqrt{3}}{2}\right)^2} \leq \frac{13}{2} \cos \theta - \frac{3\sqrt{3}}{2} \sin \theta \leq \sqrt{\left(\frac{13}{2}\right)^2 + \left(-\frac{3\sqrt{3}}{2}\right)^2}$$

$$\Rightarrow -7 \leq \frac{13}{2} \cos \theta - \frac{3\sqrt{3}}{2} \sin \theta \leq 7$$

$$\Rightarrow -7 \leq 5 \cos \theta + 3 \cos \left(\theta + \frac{\pi}{3} \right) \leq 7 \quad \text{for all } \theta.$$

$$\Rightarrow -7 + 3 \leq 5 \cos \theta + 3 \cos \left(\theta + \frac{\pi}{3} \right) + 3 \leq 7 + 3 \quad \text{for all } \theta.$$

$$\Rightarrow -4 \leq 5 \cos \theta + 3 \cos \left(\theta + \frac{\pi}{3} \right) + 3 \leq 10 \quad \text{for all } \theta.$$

Illustration 20: Find the maximum value of $1 + \sin\left(\frac{\pi}{4} + \theta\right) + 2 \cos\left(\frac{\pi}{4} - \theta\right)$ -

- (A) 1 (B) 2 (C) 3 (D) 4

Solution : We have $1 + \sin\left(\frac{\pi}{4} + \theta\right) + 2 \cos\left(\frac{\pi}{4} - \theta\right)$

$$= 1 + \frac{1}{\sqrt{2}} (\cos\theta + \sin\theta) + \sqrt{2} (\cos\theta + \sin\theta) = 1 + \left(\frac{1}{\sqrt{2}} + \sqrt{2}\right) (\cos\theta + \sin\theta)$$

$$= 1 + \left(\frac{1}{\sqrt{2}} + \sqrt{2}\right) \cdot \sqrt{2} \cos\left(\theta - \frac{\pi}{4}\right) \therefore \text{maximum value} = 1 + \left(\frac{1}{\sqrt{2}} + \sqrt{2}\right) \cdot \sqrt{2} = 4$$

Ans. (D)

Do yourself - 6 :

(i) Find maximum and minimum value of $5\cos\theta + 3\sin\left(\theta + \frac{\pi}{6}\right)$ for all real values of θ .

(ii) Find the minimum value of $\cos\theta + \cos 2\theta$ for all real values of θ .

(iii) Find maximum and minimum value of $\cos^2\theta - 6\sin\theta\cos\theta + 3\sin^2\theta + 2$.

- 6: (i) 7 & -7 (ii) $-\frac{9}{8}$ (iii) $4 + \sqrt{10}$ & $4 - \sqrt{10}$

ANSWERS FOR DO YOURSELF

1: (i) $x \in (-\infty, -2) \cup (1, 3/2)$ (ii) $x \in \mathbb{R} - (0, 1]$

2: (i) (a) $x \in (1, 8)$ (b) $x \in (-\infty, 1/2)$ (ii) $x \in (-1, 1)$

3: (i) (a) 50 (b) 4 (iv) (a) 1/2

4: (ii) 3

6: (i) 7 & -7 (ii) $-\frac{9}{8}$ (iii) $4 + \sqrt{10}$ & $4 - \sqrt{10}$

EXERCISE # 1

- The number of solution of the equation $e^{2x} + e^x + e^{-2x} + e^{-x} = 3(e^{-2x} + e^x)$ is
(A) 0 (B) 2 (C) 1 (D) more than 2
- The solution of the inequation $|x^2 - 2x - 3| < |x^2 - x + 5|$ is -
(A) $(-\infty, 5)$ (B) $(-\infty, 2) \cup (3, 8) \cup (8, \infty)$
(C) $(-8, \infty)$ (D) $(3, 8)$
- The minimum value of $f(x) = |x - 1| + |x - 2| + |x - 3|$ is equal to -
(A) 1 (B) 2 (C) 3 (D) 0
- The complete solution set of the inequation $\sqrt{x+18} < 2 - x$ is -
(A) $[-18, -2]$ (B) $(-\infty, -2) \cup (7, \infty)$ (C) $(-18, 2) \cup (7, \infty)$ (D) $[-18, -2)$
- Solution of the inequality, $x - 3 < \sqrt{x^2 + 4x - 5}$ is -
(A) $(-\infty, -5] \cup [1, \infty)$ (B) $(-5, 3]$
(C) $(-\infty, -5] \cup \left(\frac{7}{5}, \infty\right)$ (D) $\left(\frac{7}{5}, \infty\right)$
- If $\log_{0.3}(x - 1) < \log_{0.09}(x - 1)$, then x lies in the interval
(A) $(2, \infty)$ (B) $(1, 2)$ (C) $(1, \infty)$ (D) none of these
- The solution set of the inequality $\log_{\frac{5}{8}}\left(2x^2 - x - \frac{3}{8}\right) \geq 1$ is -
(A) $\left[-\frac{1}{2}, -\frac{1}{4}\right] \cup \left[\frac{3}{4}, 1\right]$ (B) $\left[-\frac{1}{2}, 1\right]$
(C) $\left[-\frac{1}{2}, \frac{1}{4}\right] \cup \left[\frac{3}{4}, 1\right]$ (D) $\left(-\infty, -\frac{1}{4}\right) \cup \left(\frac{3}{4}, \infty\right)$
- Solution set of the inequality, $2 - \log_2(x^2 + 3x) \geq 0$ is -
(A) $[-4, 1]$ (B) $[-4, -3) \cup (0, 1]$
(C) $(-\infty, -3) \cup (1, \infty)$ (D) $(-\infty, -4) \cup [1, \infty)$

9. If $\log_{1/3}\left(\frac{3x-1}{x+2}\right)$ is less than unity then x must lie in the interval -
 (A) $(-\infty, -2) \cup (5/8, \infty)$ (B) $(-2, 5/8)$
 (C) $(-\infty, -2) \cup (1/3, 5/8)$ (D) $(-2, 1/3)$
10. If a, b, c are positive real numbers such that $ab^2c^3 = 64$ then minimum value of $\left(\frac{1}{a} + \frac{2}{b} + \frac{3}{c}\right)$ is equal to:-
 (A) 6 (B) 2 (C) 3 (D) None of these
11. If $abcd = 1$ where a, b, c, d are positive reals then the minimum value of $a^2 + b^2 + c^2 + d^2 + ab + ac + ad + bc + bd + cd$ is
 (A) 6 (B) 10 (C) 12 (D) 20
12. **Statement-1:** If $27abc \geq (a+b+c)^3$ and $3a + 4b + 5c = 12$ then $\frac{1}{a^2} + \frac{1}{b^3} + \frac{1}{c^5} = 10$; where a, b, c are positive real numbers.
Statement-2: For positive real numbers $A.M. \geq G.M.$
 (A) Statement-1 is true, statement-2 is true and statement-2 is correct explanation for statement-1.
 (B) Statement-1 is true, statement-2 is true and statement-2 is NOT the correct explanation for statement-1.
 (C) Statement-1 is true, statement-2 is false.
 (D) Statement-1 is false, statement-2 is true.
13. If a, b and c are positive real numbers then $\frac{a}{b} + \frac{b}{c} + \frac{c}{a}$ is greater than or equal to
 (A) 3 (B) 6 (C) 27 (D) 5
14. If $x \in \mathbb{R}$, the numbers $(5^{1+x} + 5^{1-x})$, $a/2$, $(25^x + 25^{-x})$ form an A.P. then 'a' must lie in the interval
 (A) $[1, 5]$ (B) $[2, 5]$ (C) $[5, 12]$ (D) $[12, \infty)$

EXERCISE # 2

1. Let x, y, z be positive real numbers satisfying $x + y + z = 1$ prove that

$$xy(x + y)^2 + yz(y + z)^2 + zx(z + x)^2 \geq 4xyz$$
2. If a, b, c are positive fraction & $a + b + c = 1$, prove that $\left(1 + \frac{1}{a}\right)\left(1 + \frac{1}{b}\right)\left(1 + \frac{1}{c}\right) \geq 64$
3. Show that : $\tan A + \tan B + \tan C \geq 3\sqrt{3}$, $A, B, C \in \left[0, \frac{\pi}{2}\right]$
4. $\forall x \in \mathbb{R}$, find the range of the function, $f(x) = \cos x (\sin x + \sqrt{\sin^2 x + \sin^2 \alpha})$; $\alpha \in [0, \pi]$
5. Solve the inequality : $\log_{1/2}(x + 1) > \log_2(2 - x)$.
6. Solve the inequality : $\log_x 2 \cdot \log_{2x} 2 \cdot \log_2 4x > 1$.
7. Prove that the triangle ABC is equilateral iff, $\cot A + \cot B + \cot C = \sqrt{3}$.
8. If $A + B + C = \pi$; prove that $\tan^2 \frac{A}{2} + \tan^2 \frac{B}{2} + \tan^2 \frac{C}{2} \geq 1$.
9. If x, y, z are positive integer and $x + y + z = 1$, prove that $\left(\frac{1}{x} - 1\right)\left(\frac{1}{y} - 1\right)\left(\frac{1}{z} - 1\right) \geq 8$
10. If $2a + b + 3c = 1$ and $a > 0, b > 0, c > 0$, then find the greatest value of $a^4 b^2 c^2$ and obtain the corresponding values of a, b, c
11. If a, b, c, d are positive numbers in H.P. then prove that $a + d \geq b + c$

12. Prove that for positive quantities a, b, c we have $\frac{bc}{b+c} + \frac{ca}{c+a} + \frac{ab}{a+b} \leq \frac{a+b+c}{2}$
13. If a, b, c are natural numbers, prove that $\left(\frac{a^2+b^2+c^2}{a+b+c}\right)^{a+b+c} \geq a^a b^b c^c \geq \left(\frac{a+b+c}{3}\right)^{a+b+c}$
14. If $a \neq b$ and a, b, x, y are positive rationals such that $x + y = 1$ then prove that $a^x b^y < ax + by$
15. Find all positive real numbers x, y, z such that

$$2x - 2y + \frac{1}{z} = \frac{1}{2018}$$

$$2y - 2z + \frac{1}{x} = \frac{1}{2018}$$

$$2z - 2x + \frac{1}{y} = \frac{1}{2018}$$

Solve the following inequalities

16. $\text{Sol}(\log_2 x)^4 - \left(\log_1 \frac{x^5}{4}\right)^2 - 20\log_2 x + 148 < 0.$

17. $x^{1/\log x} \cdot \log x < 1$

18. $(\log 100x)^2 + (\log 10x)^2 + \log x \leq 14$

19. $\log_{1/5} (2x^2 + 5x + 1) < 0$

EXERCISE # 3

1. The set of all real numbers x for which $x^2 - |x + 2| + x > 0$, is [JEE 2002 (screening),3]
 (A) $(-\infty, -2) \cup (2, \infty)$ (B) $(-\infty, -\sqrt{2}) \cup (\sqrt{2}, \infty)$
 (C) $(-\infty, -1) \cup (1, \infty)$ (D) $(\sqrt{2}, \infty)$

2. If $A = \sin^2 x + \cos^4 x$, then for all real x :- [AIEEE-2011]
 (1) $1 \leq A \leq 2$ (2) $\frac{3}{4} \leq A \leq \frac{13}{16}$ (3) $\frac{3}{4} \leq A \leq 1$ (4) $\frac{13}{16} \leq A \leq 1$

3. If $a_1, a_2, a_3, \dots, a_n \in \mathbb{R}^+$ and $a_1 \cdot a_2 \cdot a_3 \dots a_n = 1$, then minimum value of $(1 + a_1 + a_1^2)(1 + a_2 + a_2^2)(1 + a_3 + a_3^2) \dots (1 + a_n + a_n^2)$ is equal to :-
 (1) 3^{n+1} (2) 3^n (3) 3^{n-1} (4) none of these [JEE-MAINS Online 2013]

4. Let x, y be positive real numbers and m, n positive integers. The maximum value of the expression $\frac{x^m y^n}{(1+x^{2m})(1+y^{2n})}$ is: [JEE-MAIN-2019]
 (1) $\frac{1}{4}$ (2) $\frac{1}{2}$ (3) 1 (4) $\frac{m+n}{6mn}$

EXERCISE # 4

MULTIPLE CORRECT CHOICE TYPE

1. A straight line through the vertex P of a triangle PQR intersects the side QR at the point S and the circumcircle of the triangle PQR at the point T. If S is not the centre of the circumcircle, then
- (A) $\frac{1}{PS} + \frac{1}{ST} < \frac{2}{\sqrt{QS \times SR}}$ (B) $\frac{1}{PS} + \frac{1}{ST} > \frac{2}{\sqrt{QS \times SR}}$
- (C) $\frac{1}{PS} + \frac{1}{ST} < \frac{4}{QR}$ (D) $\frac{1}{PS} + \frac{1}{ST} > \frac{4}{QR}$ [JEE 2008, 4]

2. If $\frac{\sin^4 x}{2} + \frac{\cos^4 x}{3} = \frac{1}{5}$ then
- (A) $\tan^2 x = \frac{2}{3}$ (B) $\frac{\sin^8 x}{8} + \frac{\cos^8 x}{27} = \frac{1}{125}$
- (C) $\tan^2 x = \frac{1}{3}$ (D) $\frac{\sin^8 x}{8} + \frac{\cos^8 x}{27} = \frac{2}{125}$ [JEE-2009-4]

3. Let $b_i > 1$ for $i = 1, 2, \dots, 101$. Suppose $\log_e b_1, \log_e b_2, \dots, \log_e b_{101}$ are in Arithmetic progression (A.P.) with the common difference $\log_e 2$. Suppose $a_1, a_2, \dots, a_{101}$ are in A.P. such that $a_1 = b_1$ and $a_{51} = b_{51}$. If $t = b_1 + b_2 + \dots + b_{51}$ and $S = a_1 + a_2 + \dots + a_{51}$ then [IIT-JEE 2016, Paper-2]
- (A) $s > t$ and $a_{101} > b_{101}$ (B) $s > t$ and $a_{101} < b_{101}$
- (C) $s < t$ and $a_{101} > b_{101}$ (D) $s < t$ and $a_{101} < b_{101}$

Comprehension (3 questions)

Let A_1, G_1, H_1 denote the arithmetic, geometric and harmonic means, respectively, of two distinct positive numbers. For $n \geq 2$, let A_{n-1} and H_{n-1} have arithmetic, geometric and harmonic means as A_n, G_n, H_n respectively.

4. Which one of the following statements is correct?
- (A) $G_1 > G_2 > G_3 > \dots$
- (B) $G_1 < G_2 < G_3 < \dots$
- (C) $G_1 = G_2 = G_3 = \dots$
- (D) $G_1 < G_3 < G_5 < \dots$ and $G_2 > G_4 > G_6 > \dots$ [JEE 2007, 4]
5. Which one of the following statements is correct?
- (A) $A_1 > A_2 > A_3 > \dots$
- (B) $A_1 < A_2 < A_3 < \dots$
- (C) $A_1 > A_3 > A_5 > \dots$ and $A_2 < A_4 < A_6 < \dots$
- (D) $A_1 < A_3 < A_5 < \dots$ and $A_2 > A_4 > A_6 > \dots$ [JEE 2007, 4]
6. Which one of the following statements is correct?
- (A) $H_1 > H_2 > H_3 > \dots$
- (B) $H_1 < H_2 < H_3 < \dots$
- (C) $H_1 > H_3 > H_5 > \dots$ and $H_2 < H_4 < H_6 < \dots$
- (D) $H_1 < H_3 < H_5 < \dots$ and $H_2 > H_4 > H_6 > \dots$ [JEE 2007, 4]

7. The minimum value of the sum of real numbers $a^{-5}, a^{-4}, 3a^{-3}, 1, a^8$ and a^{10} with $a > 0$ is

EXERCISE # 5

1. Which of the following statements is(are) correct ?
 (A) $7^{1/7} > (42)^{1/14} > 1$ (B) $\log_3(5) \log_7(9) \log_{11}(13) > -2$
 (C) $\sqrt{99+70\sqrt{2}} + \sqrt{99-70\sqrt{2}}$ is rational (D) $\frac{1}{\log_4 3} + \frac{1}{\log_7 3} > 3$

MATCH THE COLUMN

2. Consider the function $f(x) = |x - 1| - 2|x + 2| + |x + 3|$

Column-I

- (A) If $f(x) = k$ has no solution, then $k \in$
 (B) If $f(x) = k$ has one solution, then $k \in$
 (C) If $f(x) = k$ has two solution, then $k \in$
 (D) If $f(x) = k$ has more then two solution, then $k \in$

Column-II

- (p) (2, 4)
 (q) $(-\infty, -2) \cup (4, \infty)$
 (r) $(-2, 2) \cup \{4\}$
 (s) $\{-2, 2\}$

3. If $f(x) = \frac{\sin 3x}{\sin x}$, $x \neq n\pi$, then the range of values of $f(x)$ for real values of x is -

- (A) $[-1, 3]$ (B) $(-\infty, -1]$ (C) $(3, +\infty)$ (D) $[-1, 3)$

4. Maximum and minimum value of $2\sin^2\theta - 3\sin\theta + 2$ is -

- (A) $\frac{1}{4}, -\frac{7}{4}$ (B) $\frac{1}{4}, \frac{21}{4}$ (C) $\frac{21}{4}, -\frac{3}{4}$ (D) $7, \frac{7}{8}$

5. For $\theta \in (0, \pi/2)$, the maximum value of $\sin\left(\theta + \frac{\pi}{6}\right) + \cos\left(\theta + \frac{\pi}{6}\right)$ is attained at $\theta =$

- (A) $\frac{\pi}{12}$ (B) $\frac{\pi}{6}$ (C) $\frac{\pi}{3}$ (D) $\frac{\pi}{4}$

6. Minimum value of the expression $\cos^2\theta - (6 \sin \theta \cos \theta) + 3 \sin^2\theta + 2$, is -

- (A) $4 + \sqrt{10}$ (B) $4 - \sqrt{10}$ (C) 0 (D) 4

7. $\sec^2\theta = \frac{4xy}{(x+y)^2}$ is true if and only if-

- (A) $x + y \neq 0$ (B) $x = y, x \neq 0$ (C) $x = y$ (D) $x \neq 0, y \neq 0$

EXERCISE # 6

- If $a > 0$, then minimum value of $a + 2a^2 + a^3 + 15 + a^{-1} + a^{-3} + a^{-4}$ is
- If $a, b, c, d > 0$ such that $a + 2b + 3c + 4d = 50$, then find the maximum value of $\left(\frac{a^2b^4c^3d}{16}\right)^{1/10}$.
- Given that the cubic $ax^3 - ax^2 + 9bx - b = 0$ ($a \neq 0$) has all three positive roots. Find the harmonic mean of the roots independent of a and b , hence deduce that the roots are all equal. Find also the minimum value of $(a + b)$ if a and $b \in \mathbb{N}$.
- If $y = 10 \cos^2 x - 6 \sin x \cos x + 2 \sin^2 x$, then find the greatest & least value of y .
 - If $y = 1 + 2 \sin x + 3 \cos^2 x$, find the maximum & minimum values of $y \forall x \in \mathbb{R}$.
 - If $y = 9 \sec^2 x + 16 \operatorname{cosec}^2 x$, find the minimum value of y for all permissible value of x .
 - If $a \leq 3 \cos\left(\theta + \frac{\pi}{3}\right) + 5 \cos \theta + 3 \leq b$, find a and b , where a is the minimum value & b is the maximum value.
- If M and m denotes maximum and minimum value of $\sqrt{49 \cos^2 \theta + \sin^2 \theta} + \sqrt{49 \sin^2 \theta + \cos^2 \theta}$ then find the value of $(M + m)$.
- If x and y are real number such that $x^2 + 2xy - y^2 = 6$, find the minimum value of $(x^2 + y^2)^2$.
- Let α, β, γ be distinct real numbers such that

$$a\alpha^2 + b\alpha + c = (\sin \theta)\alpha^2 + (\cos \theta)\alpha$$

$$a\beta^2 + b\beta + c = (\sin \theta)\beta^2 + (\cos \theta)\beta$$

$$a\gamma^2 + b\gamma + c = (\sin \theta)\gamma^2 + (\cos \theta)\gamma$$
 (where $a, b, c \in \mathbb{R}$)
 - Find the maximum value of the expression $\frac{a^2 + b^2}{a^2 + 3ab + 5b^2}$
 - If $\vec{V}_1 = a\hat{i} + b\hat{j} + c\hat{k}$ makes an angle $\frac{\pi}{3}$ with $\vec{V}_2 = \hat{i} + \hat{j} + \sqrt{2}\hat{k}$, then find the number of values of $\theta \in [0, 2\pi]$

8. Given $x, y \in \mathbb{R}$, $x^2 + y^2 > 0$. If the maximum and minimum value of the expression

$$E = \frac{x^2 + y^2}{x^2 + xy + 4y^2}$$
 are M and m , and A denotes the average value of M and m , compute

(2007)A.

Solve the following inequalities

9. $\log_{1/2} x + \log_3 x > 1$

10. $\log_{x^2}(2+x) < 1$

11. $\log_x \frac{4x+5}{6-5x} < -1$

12. $(\log_{|x+6|} 2) \cdot \log_2(x^2 - x - 2) \geq 1$

13. $\log_3 \frac{|x^2 - 4x| + 3}{x^2 + |x - 5|} \geq 0$

14. $\log_{[(x+6)/3]} [\log_2 \{(x-1)/(2+x)\}] > 0$

ANSWERS KEY

EXERCISE # 1

1. C 2. C 3. B 4. D 5. A 6. A 7. A
8. B 9. A 10. C 11. B 12. D 13. A 14. D

EXERCISE # 2

4. $-\sqrt{1+\sin \alpha} \leq y \leq \sqrt{1+\sin^2 \alpha}$ 5. $-1 < x < \frac{1-\sqrt{5}}{2}$ or $\frac{1+\sqrt{5}}{2} < x < 2$
6. $2^{-\sqrt{2}} < x < 2^{-1}; 1 < x < 2^{\sqrt{2}}$ 10. $\frac{1}{9.2^{16}}; a = \frac{1}{4}, b = \frac{1}{4}, c = \frac{1}{12}$
15. $x = y = z = 2018$ 16. $x \in \left(\frac{1}{16}, \frac{1}{8}\right) \cup (8, 16)$ 17. $(0, 1) \cup (1, 10^{1/10})$
18. $\frac{1}{\sqrt{10}^9} \leq x \leq 10$ 19. $(-\infty, -2.5) \cup (0, \infty)$

EXERCISE # 3

1. B 2. 3 3. B 4. 1

EXERCISE # 4

1. B,D 2. A,B 3. B 4. C 5. A 6. B 7. 8

EXERCISE # 5

1. A,B,D 2. $(A) \rightarrow q; (B) \rightarrow r; (C) \rightarrow p; (D) \rightarrow s$ 3. D 4. D
5. A 6. B 7. B

EXERCISE # 6

1. 22 2. 5 3. 28
4. (a) $y_{\max} = 11$, $y_{\min} = 1$; (b) $y_{\max} = \frac{13}{3}$, $y_{\min} = -1$; (c) 49; (d) $a = -4$ & $b = 10$
5. 18 6. 18 7. (a) 2, (b) 3 8. 1338
9. $0 < x < 3^{1/1-\log 3}$ (where base of log is 2)
10. $-2 < x < -1$, $-1 < x < 0$, $0 < x < 1$, $x > 2$ 11. $\frac{1}{2} < x < 1$
12. $x < -7$, $-5 < x \leq -2$, $x \geq 4$ 13. $x \leq -\frac{2}{3}$; $\frac{1}{2} \leq x \leq 2$
14. $(-6, -5) \cup (-3, -2)$

